

Mauricio Angel Treviño IV

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EDUCATION

Rice University, Houston, TX

August 2025 – Present

Master of Science in Space Studies, Expected Graduation December 2026

GPA: 4.0/4.0

Current Coursework: Computational Fluid Mechanics, Solar System Physics, Data Science for Engineering Leaders

Texas A&M University, College Station, TX

August 2020 – May 2025

Bachelor of Science in Mechanical Engineering, Engineering Honors, Summa Cum Laude

GPA: 3.9/4.0

TECHNICAL SKILLS

- **CAD:** SolidWorks, CREO, Fusion 360 | **FEA:** SolidWorks Simulation, ANSYS
- **CFD:** ANSYS Fluent | **Simulation & Analysis:** MATLAB, FloMaster, LabVIEW | **Programming:** Python
- **Manufacturing:** CNC Mill (MasterCAM), FDM 3D Printing, Manual Tools | **Circuit Design:** Multisim

PROFESSIONAL EXPERIENCE

Society of Automotive Engineers Formula Internal Combustion, College Station, TX

May 2024 – May 2025

Aerodynamics Sub-Team Member

- Tuned event-specific setups using simulation and track validation, contributing to 2nd place overall at the 2025 FSAE Michigan competition
- Led aerodynamic development through full-car CFD simulations in Ansys Fluent, optimizing interactions between wings, undertray, and bodywork for performance gains
- Directed cross-subteam collaboration between aerodynamics, suspension, powertrain, and chassis to ensure cohesive vehicle integration and maximize overall performance
- Manufactured the full aerodynamics package using carbon fiber epoxy composites

Robotics and Automation Design (RAD) Lab, College Station, TX

May 2024 – July 2025

Undergraduate Research Assistant

- Engineered a deployable sensor package launched from RoboBall III to extend mission range, provide GPS localization, and enable visual reconnaissance
- Developed and integrated key subsystems for RoboBall II, including a wireless inductive charging module, real-time data visualization via ROS2/MATLAB, and explored soft-shelled mobility platforms for diverse environments
- Co-authored 1 published work, viewable on my ORCID page: [0009-0002-2986-3486](https://orcid.org/0009-0002-2986-3486)

NASA Pathways Co-Op, Stennis Space Center, MS

January 2023 - August 2023

Rocket Propulsion Test Engineer Co-Op

January 2024 - May 2024

- Designed and analyzed cryogenic piping and instrumentation systems for rocket propulsion testing, adhering to ASME standards and NASA protocols
- Conducted liquid oxygen transfers and supported test stand integration using model-based engineering approaches
- Evaluated novel valve performance, determining flow coefficients and assessing system-level impacts
- Selected as NASA PAXC Stennis Representative; presented technical work at the National Intern Symposium

Society of Automotive Engineers Aerospace Design, College Station, TX

April 2021 – May 2024

Product Owner of Structures and Material Science Sub-Team

- Optimized the internal truss structure of an RC aircraft using FEA and hand calculations
- Constructed the aircraft using precision fabrication techniques including custom jig assemblies
- Led the Micro Class team to 1st place overall at the 2024 SAE Aero Design West Competition, applying Agile methods to streamline development and reduce downtime

CERTIFICATIONS & INTERESTS

Certifications: Americas Cutting Edge (ACE) Manufacturing Certificate, HAAS Basic Mill and Lathe Certificates, Solidworks CAD Design Associate Certificate

Interests: Extraterrestrial Exploration, Lander Systems, Propulsion, Spikeball, FPV Drones